

Linear regression – Maths

PART – 1

$$d(x^n) = n * x^{n-1}$$

$$y = x^2$$

$$\frac{dy}{dx} = \frac{dx^2}{dx} = y' = 2 * x^{2-1} = 2x$$

$$y = x^2 + x$$

$$y' = 2x + 1$$

How to find any minimum point of equation or curve

- *Minimum point means there is No slope*
- *slope mean differentiation*
- *so in order to find the minimum point first find the slope*
- *means first find the differentiation*
- *and make that slope = 0*
- *then we can able to find the minimum point*

example – 1: $y = x^2$

$$y' = 2x$$

$$2x = 0$$

$$x = 0$$

so $y = x^2$ has minimum point at $x = 0$

example – 2: $y = x^2 + 2x$

$$y' = 2x + 2$$

$$2x + 2 = 0$$

$$x = -1$$

differentiation is nothing but find the slope

$$\frac{dy}{dx} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$y = (x^2 + 1)^2$$

$$y = 2 * (x^2 + 1) * \left[\frac{dx^2}{dx} + \frac{d(1)}{dx} \right] = 2 * (x^2 + 1) * [2x + 0] = 2 * (x^2 + 1) * 2x$$

$$y = (3x^2 + x + 1)^2 = 2 * (3x^2 + x + 1) \left[\frac{d3x^2}{dx} + \frac{d(x)}{dx} + \frac{d(1)}{dx} \right]$$

$$2 * (3x^2 + x + 1) * (6x + 1)$$

PART - 2 :

Partial differentiation

$x^2 + y$ combination of x and y

we need to differentiate w.r.t x and w.r.t y

partially with x and partially with y

$$f(x) = x^2 \quad f(y) = y \quad f(x, y) = x^2 + y \quad f(x, y, z) = x^2 + y + 2z$$

$$\frac{\partial f(x, y)}{\partial x}, \quad \frac{\partial f(x, y)}{\partial y}$$

$$\frac{\partial (x^2 + y)}{\partial x} = \frac{\partial x^2}{\partial x} + \frac{\partial y}{\partial x} = 2x + 0 = 2x$$

$$\frac{\partial (x^2 + y)}{\partial y} = \frac{\partial x^2}{\partial y} + \frac{\partial y}{\partial y} = 0 + 1 = 1$$

$$f(x, y) = x^2 + y^2 \quad g(x, y) = x + y - 1$$

$$L(x, y, \lambda) = f(x, y) - \lambda * g(x, y)$$

$$L(x, y, \lambda) = x^2 + y^2 - \lambda * [x + y - 1]$$

$$L(x, y, \lambda) = x^2 + y^2 - \lambda x - \lambda y + \lambda$$

$$\frac{\partial L}{\partial x} = \frac{\partial (x^2 + y^2 - \lambda x - \lambda y + \lambda)}{\partial x} = 2x - \lambda$$

$$\frac{\partial L}{\partial y} = \frac{\partial (x^2 + y^2 - \lambda x - \lambda y + \lambda)}{\partial y} = 2y - \lambda$$

$$\frac{\partial L}{\partial \lambda} = \frac{\partial (x^2 + y^2 - \lambda x - \lambda y + \lambda)}{\partial \lambda} = -x - y + 1$$

$$\frac{\partial L}{\partial x}, \frac{\partial L}{\partial y}, \frac{\partial L}{\partial \lambda}$$

PART – 3:

Find the minimum point of the equation

$$y = x^2 \quad \text{what is minimum point}$$

minimum point means, if you substitute the point, the entire equation = 0

minimum point means, at that point slope = 0

In maths if you want to find minimum point

- *find the slope of the equation*
- *which means find the differentiation*
- *make that differentiation equals to zero*
- *That is your minimum point*

$$y = x^2$$

$$y' = 2x \implies 2x = 0 \implies x = 0$$

$$y = x^2 + x$$

$$y' = 2x + 1$$

$$2x + 1 = 0, \quad x = -\frac{1}{2}$$

$$y = x^2 + x \text{ has minimum point at } x = -\frac{1}{2}$$